

Calibrated Agents as Market Makers: Automated Liquidity Provision for Long-Tail Prediction Markets

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Abstract—Under what conditions can a forecasting agent profitably provide liquidity on a prediction market? We derive a decomposition of the optimal market-making spread into a microstructure term and an adverse selection term proportional to the agent’s mean absolute estimation error α . The profitability condition $s > 2\varphi\alpha$ depends only on the sign of per-fill profit and holds for any positive fill-rate function. A threshold theorem shows that tolerable calibration error increases with existing market spread—a comparative static suggesting that calibrated agents may add the most value on illiquid long-tail markets that lack human market makers. Monte Carlo simulations across 3,000 synthetic markets calibrated to empirical Polymarket parameters ($\kappa = 36$) show win rates near 68–70% across calibration levels, with mean profit varying by $3.4\times$ (\$5.83 to \$1.72), confirming that calibration quality drives profit magnitude. On the empirical side, the main finding is a calibration gap: five frontier LLMs evaluated on 15 resolved Polymarket questions reveal that market-alignment error (MAD ≤ 0.029) underestimates true estimation error (α up to 0.437) by an order of magnitude—a result with implications for any system using market prices as calibration targets. The model omits inventory risk and multi-agent competition, limiting direct deployment conclusions, but the spread decomposition provides a tractable framework for analyzing when AI-agent liquidity provision is viable.

Index Terms—prediction markets, automated market making, LLM calibration, mechanism design, DeFi liquidity, adverse selection, Glosten–Milgrom

I. INTRODUCTION

Prediction markets aggregate dispersed information into probability estimates [1], [2], but suffer from chronic illiquidity on niche questions. An analysis of 290,000 Polymarket contracts reveals extreme concentration: 63% of active short-term markets have zero 24-hour volume [3]. Thin markets produce wide spreads and unreliable probabilities, defeating the core purpose of a prediction market.

Meanwhile, LLM-based forecasting has progressed rapidly. Systems have improved from crowd-level accuracy (Brier ≈ 0.121) in 2023 to near-superforecaster performance (Brier ≈ 0.101) by late 2025 [4], [5]. RL-tuned models achieve ECE of 0.042 [6], surpassing frontier models in calibration at a cost of fractions of a cent per estimate.

On-chain prediction markets such as Polymarket (deployed on Polygon) operate as decentralized limit-order-book venues where anyone can post liquidity autonomously. Unlike centralized venues, there is no designated market maker, making

illiquidity a structural problem. A forecasting agent producing calibrated probability estimates at near-zero cost could fill this gap by posting bid-ask quotes derived from its forecasts.

This paper treats calibration error α as a key economic input—jointly with the informed trader fraction φ and fill-rate sensitivity κ —and derives conditions under which agent-provided liquidity is profitable.

Contributions.

- 1) **Spread decomposition.** The optimal spread $s^* = 1/\kappa + 2\varphi\alpha$ separates a microstructure term from an adverse selection term. The profitability condition $s > 2\varphi\alpha$ holds for any positive fill-rate function (Sections III–IV).
- 2) **Long-tail comparative static.** A threshold theorem shows that tolerable calibration error increases with existing market spread: agents can improve wider-spread markets even with worse calibration. Under this model (no inventory risk, static φ), this reverses the direction of marginal value relative to traditional market making (Section IV).
- 3) **Monte Carlo validation** across 3,000 synthetic markets calibrated to empirical $\kappa = 36$, showing that optimal spreads stabilize win rates near 68–70% while profit varies by $3.4\times$ across calibration levels (Section V).
- 4) **Calibration-gap finding:** a resolved-outcome backtest on 15 questions reveals that market-alignment error (MAD) underestimates true estimation error α by an order of magnitude—the paper’s most practically relevant empirical result (Section VI).

II. BACKGROUND AND RELATED WORK

Prediction markets and automated market making. The LMSR [7] is the most influential AMM for prediction markets; Othman et al. [8] demonstrated practical viability and Chen and Pennock [9] provided a utility-based foundation. These are *uninformed* market makers that provide liquidity through a mechanistic cost function with bounded worst-case loss. Our agent market maker is *informed*—its profitability depends explicitly on calibration quality. The Glosten–Milgrom model [10] decomposes the spread into adverse selection and inventory costs; this paper adapts that framework to prediction markets with calibrated agents. DeFi AMMs [11] face binary outcome risk on prediction markets (shares resolve to 0 or 1), making the informed-agent approach qualitatively different.

LLM forecasting and calibration.: LLM forecasting has progressed from Brier ≈ 0.179 [4] to near-superforecaster performance (≈ 0.108) [5]. Turtel [6] achieved ECE of 0.042 via RL with proper scoring rule rewards on a 14B-parameter model. Chandak et al. [12] showed calibration gains transfer to out-of-distribution benchmarks. Schoenegger et al. [13] demonstrated 12-LLM ensembles match human crowd accuracy. Yang et al. [14] survey LLM forecasting capabilities and emphasize the remaining calibration gap. The cost per estimate has fallen from hundreds of dollars (human superforecasters) to fractions of a cent (RL-tuned open models).

Calibration in decision-making and related frameworks.: Wright [15] analyzes utility-weighted calibration under trading frictions in a portfolio context, but does not consider prediction markets. Todasco [16] uses a simulated prediction market to elicit LLM confidence, but does not derive optimal spreads. Page [17] showed that prediction-market prices become miscalibrated for long-dated events. The equivalence between market-makers and online learning algorithms [18] and stochastic kernel approaches to prediction market pricing [19] are related but orthogonal to the present contribution. None of these works derives an optimal bid-ask spread from the forecaster's calibration error.

III. MODEL

A. Market Environment

Definition III.1 (Binary Prediction Market). A binary prediction market $\mathcal{M} = (E, T, \mathcal{O})$ consists of:

- (i) An event E with binary outcome $o \in \{0, 1\}$ and true probability $p^* = \Pr(o = 1)$;
- (ii) A trading period $[0, T]$ before resolution;
- (iii) An order book $\mathcal{O}$ with bids and asks from human participants.

The market's existing effective spread is s_m , reflecting the cost of immediate execution.

B. Agent Market Maker

Definition III.2 (Agent Market Maker). An agent market maker $\mathcal{A} = (\mathcal{F}, s, q)$ consists of:

- (i) A forecasting model $\mathcal{F} : \text{context} \rightarrow [0, 1]$ producing probability estimate $\hat{p}$;
- (ii) A spread $s > 0$ determining quote width;
- (iii) A quantity $q > 0$ determining order size.

The agent posts bid $b = \hat{p} - s/2$ and ask $a = \hat{p} + s/2$.

Assumption III.3 (Estimation Error Model). The agent's estimate satisfies $\hat{p} = p^* + \varepsilon$, where ε is a zero-mean noise term. Define the mean absolute estimation error (MAEE):

$$\alpha \equiv \mathbb{E}[|\hat{p} - p^*|] = \mathbb{E}[|\varepsilon|]. \quad (1)$$

Conservative under bias.: Table II shows all models exhibit small negative bias ($\mathbb{E}[\varepsilon] \in [-0.002, -0.023]$). When $\mathbb{E}[\varepsilon] \neq 0$, the expected adverse selection cost becomes $\varphi \cdot \mathbb{E}[|\varepsilon|]$, where $\mathbb{E}[|\varepsilon|] \geq |\mathbb{E}[\varepsilon]|$ by Jensen's inequality. Hence the zero-mean assumption is *conservative*: it underestimates the true adverse selection loss. The profitability condition $s > 2\varphi\alpha$

remains sufficient but the realized loss may exceed $\varphi\alpha$ under systematic bias.

Lemma III.4 (ECE as Lower Bound on MAEE). For any forecasting system evaluated with B -bin expected calibration error: $\text{ECE}_B \leq \alpha$, with equality in the limit $B \rightarrow \infty$. This is a direct application of Jensen's inequality within each calibration bin; it is well known but included for completeness.

Proof. Jensen's inequality applied to $|\cdot|$ within each bin. See Appendix A for details. $\square$

C. Order Arrival Process

Assumption III.5 (Trader Model). Orders arrive from three sources: (i) informed traders at rate λ_I who observe p^* and select the side unfavorable to the agent (standard Glosten–Milgrom adverse selection); (ii) uninformed traders at rate λ_U who trade randomly; (iii) noise traders at rate λ_N . Define $\varphi = \lambda_I/(\lambda_I + \lambda_U + \lambda_N) \in [0, 1]$.

Assumption III.6 (Fill Rate). The aggregate fill rate decreases exponentially with spread:

$$F(s) = \lambda \exp(-\kappa s), \quad (2)$$

where $\lambda = \lambda_I + \lambda_U + \lambda_N$ and $\kappa > 0$ governs price sensitivity.

IV. MAIN RESULTS

A. Expected Profit

Proposition IV.1 (Profit per Fill). When the agent's bid at $b = \hat{p} - s/2$ is filled, expected profit is $\pi_{\text{bid}} = s/2 + (p^* - \hat{p})$. When the ask is filled: $\pi_{\text{ask}} = s/2 - (p^* - \hat{p})$.

Under Assumption III.5, informed traders select the unfavorable side, costing the agent $s/2 - |\hat{p} - p^*|$ per fill. Uninformed traders earn $s/2$ for the agent in expectation.

Theorem IV.2 (Expected Profit Rate). Under Assumptions III.3–III.6, the agent's expected profit per unit time is:

$$\mathbb{E}[\Pi(s)] = F(s) \cdot \left[\frac{s}{2} - \varphi \cdot \alpha \right]. \quad (3)$$

Proof. Per fill, the agent earns $s/2 - \varphi|\hat{p} - p^*|$ in expectation. Multiplying by $F(s)$ and taking expectations gives (3). See Appendix A. $\square$

Corollary IV.3 (Profitability Condition). The agent is profitable in expectation iff $s > 2\varphi \cdot \alpha$.

The profitability condition is conservative: in a refined model where informed traders only trade when $|\hat{p} - p^*| > s/2$, the spread screens out marginal informed traders, making the condition sufficient but not necessary (Appendix A).

B. Optimal Spread

Theorem IV.4 (Optimal Spread). Under the exponential fill rate (2), the spread maximizing expected profit is:

$$s^* = \frac{1}{\kappa} + 2\varphi\alpha. \quad (4)$$

The maximum expected profit rate is $\Pi^* = (\lambda/\kappa) \cdot \exp(-1 - 2\kappa\varphi\alpha)$.

Proof. Maximizing $\mathbb{E}[\Pi(s)] = \lambda e^{-\kappa s}(s/2 - \varphi\alpha)$ yields $s^* = 1/\kappa + 2\varphi\alpha$. See Appendix A. $\square$

The optimal spread decomposes into a *microstructure* term $1/\kappa$ (independent of estimation quality) and an *adverse selection* term $2\varphi\alpha$ (proportional to estimation error).

C. Calibration Threshold for Market Quality Improvement

Theorem IV.5 (Estimation Error Threshold). *The agent improves market quality (tighter effective spread) on a market with spread s_m iff:*

$$\alpha < \alpha^* \equiv \frac{s_m - 1/\kappa}{2\varphi}. \quad (5)$$

Corollary IV.6 (Long-Tail Advantage). *α^* is strictly increasing in s_m : $\partial\alpha^*/\partial s_m = 1/(2\varphi) > 0$. The tolerable calibration error for market improvement is higher on wider-spread markets. Traditional market makers avoid illiquid venues due to inventory risk and thin volume; an informed agent with negligible per-query cost faces neither constraint, reversing the direction of marginal value under this model’s assumptions.*

D. Robustness Under Endogenous Informed Fraction

Proposition IV.7 (Monotonicity Under Rising φ). *Let $\varphi(t)$ be non-decreasing. Then: (i) $s^*(t)$ is non-decreasing; (ii) $\Pi^*(t)$ is non-increasing; (iii) there exists $\varphi_{\max} = (s_m - 1/\kappa)/(2\alpha)$ beyond which the agent can no longer improve the market.*

Proof. Parts (i)–(ii) follow from monotonicity of s^* and Π^* in φ . Part (iii) follows from $s^* = s_m$. $\square$

On thin markets where $s_m \gg 1/\kappa$, the critical $\varphi_{\max}$ is large, sustaining profitability even under substantial informed-trader entry. A full equilibrium analysis of multi-agent entry dynamics is left to future work.

V. SIMULATION EXPERIMENTS

A. Simulation Design

Each of 3,000 simulated markets has true probability $p^* \sim U(0.05, 0.95)$, agent estimate $\hat{p} = \text{clip}(p^* + \varepsilon, 0.01, 0.99)$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, a potential pool of 2,000 trader arrivals filtered through the fill-rate function $F(s) = \exp(-\kappa s)$ with $\hat{\kappa} = 36$, $\varphi = 0.15$, and Bernoulli resolution. Each agent trades at its optimal spread $s^*(\alpha)$. Three α levels are tested: 0.042 (RL-tuned; [6]), 0.10 (frontier + retrieval; [5]), and 0.17 (legacy; [4]).

B. Key Results

Fig. 1 presents the core mechanism. Fig. 2 shows Monte Carlo results calibrated to the empirical $\hat{\kappa} = 36$. These simulations verify internal consistency of the model; they do not constitute independent validation, since the data-generating process matches the assumptions.

Because each agent trades at its optimal spread $s^*(\alpha)$, win rates are stabilized near 68–70%: the spread formula absorbs calibration differences into quote width. The economic impact of calibration instead manifests in *profit magnitude*: the RL-tuned agent ($\alpha = 0.042$) achieves mean P&L of \$5.83 per

TABLE I
SENSITIVITY TO κ ($\alpha = 0.042$, $\varphi = 0.15$).

	$\kappa = 20$	$\kappa = 36$	$\kappa = 60$
$1/\kappa$ (microstructure)	5.0%	2.8%	1.7%
s^* (optimal spread)	6.3%	4.0%	2.9%
Profitability threshold	$s > 1.3\%$ (invariant)		

market (win rate $69.7\% \pm 1.6\%$), versus \$3.60 for frontier ($\alpha = 0.10$) and \$1.72 for legacy ($\alpha = 0.17$)—a $3.4\times$ ratio. Fill volume also varies substantially (313 vs. 243 vs. 115 average fills), reflecting the tighter spreads enabled by lower α .

Empirical calibration of κ from 98 active Polymarket markets yields $\hat{\kappa} \approx 36$ via nonlinear least squares, implying $1/\kappa \approx 2.8\%$ —small relative to typical thin-market spreads. As Fig. 6(c) shows, implied κ varies from ≈ 20 (low-volume, Q1) to ≈ 59 (high-volume, Q4–Q5). Table I reports key results across this range: the profitability threshold $s > 2\varphi\alpha$ is independent of κ , but the optimal spread and microstructure term scale as $1/\kappa$.

Robustness to fill-rate model.: The profitability condition $s > 2\varphi\alpha$ holds for any positive fill-rate $F(s) > 0$: since $\mathbb{E}[\Pi(s)] = F(s) \cdot [s/2 - \varphi\alpha]$, the sign depends only on the per-fill profit $s/2 - \varphi\alpha$. This invariance is structural rather than coincidental. The *optimal spread* does depend on the fill-rate specification: under linear fill-rate $F(s) = \lambda \max(0, 1 - \kappa s)$, $s^* = 1/(2\kappa) + \varphi\alpha$; under power-law $F(s) = \lambda/(1 + \kappa s)^2$, $s^* = 1/\kappa + 2\varphi\alpha$; identical to the exponential case. All three preserve the additive microstructure + adverse selection decomposition, though with different coefficients.

Stress scenarios.: Two stress scenarios were simulated. (i) *Calibration drift*: α increases from 0.042 to 0.15 over 1,000 markets while the agent continues to trade at a fixed spread. Win rate degrades smoothly from $\sim 70\%$ toward $\sim 60\%$, with cumulative P&L remaining positive. (ii) *Informed-fraction surge*: φ doubles from 0.15 to 0.30 mid-simulation; profitability declines, consistent with Proposition IV.7. Both confirm graceful degradation.

Full sensitivity analysis, parameter calibration, and market quality impact figures are in the supplementary material (Appendix A).

VI. EMPIRICAL VALIDATION

A. Experimental Design

Five frontier LLMs (Claude Opus/Sonnet 4.6, GPT-5.2 Pro/5.2, Grok 4.1) were queried via OpenRouter on February 18, 2026 on 12 active Polymarket markets spanning four domains, producing 60 forecasts with identical prompts (temperature 0.2).

B. Model–Market Alignment

Table II presents alignment metrics. All five models achieve high correlation with market prices ($r > 0.988$; Fisher- z 95% CI at $n = 12$: $[0.95, 0.998]$). However, high correlation with market prices is expected—LLMs and markets draw on similar

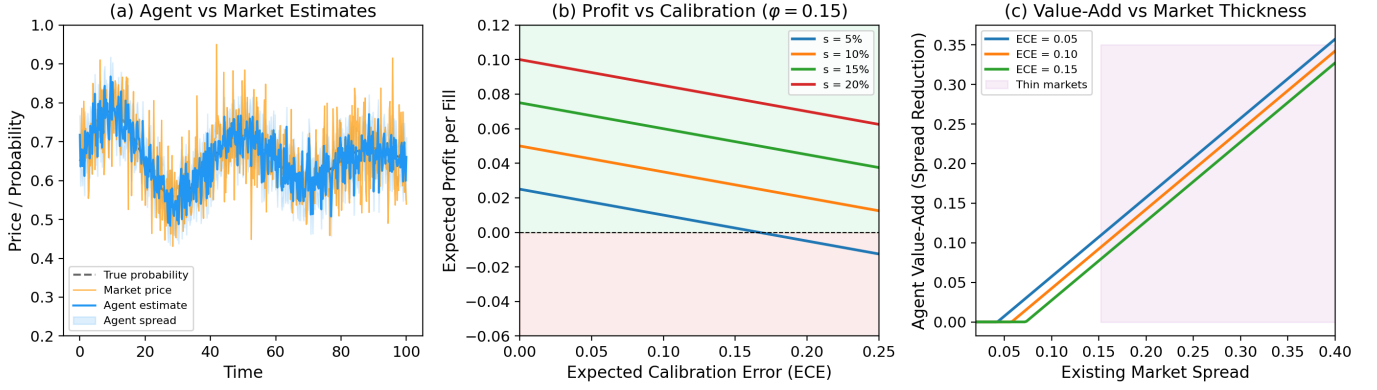


Fig. 1. Core Mechanism. (a) Agent estimate tracks true probability. (b) Expected profit per fill vs. α for different spreads. (c) Agent value-add vs. existing market spread.

TABLE II
MODEL COMPARISON ON 12 LIVE POLYMARKET MARKETS. MAD:
MEAN ABSOLUTE DEVIATION. BIAS: MEAN SIGNED DEVIATION.

Model	MAD	Corr.	Bias	Lat.
Cl. Opus 4.6	0.017	0.995	−0.015	13.6s
Cl. Sonnet 4.6	0.019	0.993	−0.012	10.4s
GPT-5.2 Pro	0.017	0.996	−0.012	42.9s
GPT-5.2	0.029	0.988	−0.023	15.1s
Grok 4.1	0.015	0.996	−0.002	8.3s
Ensemble	0.018	0.995	−0.013	—

TABLE III
RESOLVED-OUTCOME BACKTEST (15 QUESTIONS). α = TRUE MEAN
ABSOLUTE ESTIMATION ERROR.

Model	Brier	ECE	α	Acc.
Cl. Opus 4.6	0.071	0.113	0.149	93%
Cl. Sonnet 4.6	0.076	0.123	0.147	87%
GPT-5.2 Pro	0.309	0.261	0.427	60%
GPT-5.2	0.249	0.097	0.437	53%
Grok 4.1	0.222	0.225	0.281	73%

public information—and does not establish independent predictive power. The MAD range is 1.5–2.9%, with systematic negative bias (−0.2% to −2.3%). These are *alignment* metrics, not calibration: true α requires resolved outcomes.

C. Resolved-Outcome Evaluation

To provide true calibration metrics, the same five models were evaluated on 15 resolved Polymarket questions with known binary outcomes. Training-data contamination is a concern for well-known events (Appendix G).

Models that appeared nearly identical in market alignment (MAD = 0.015–0.029) show dramatically different true estimation errors (α = 0.147–0.437), confirming that market-alignment error is a poor proxy for α (Lemma III.4). Bootstrap resampling (10,000 iterations) yields 95% CIs for the best model: $\alpha \in [0.09, 0.22]$ for Claude Opus 4.6. The sample size ($n = 15$) limits precision; expanding to $n \geq 50$ would narrow CIs by $\sim 40\%$.

GPT-5.2 Pro anomaly.: GPT-5.2 Pro shows worse true calibration ($\alpha = 0.427$) than GPT-5.2 base ($\alpha = 0.437$) despite superior market alignment (MAD 0.017 vs. 0.029). The Pro model produces more decisive (extreme) probability estimates; when these confident predictions are wrong, they incur large absolute errors. This illustrates the insufficiency of market-alignment metrics: high confidence improves alignment when correct but amplifies α when incorrect.

Training-data contamination.: For well-known resolved events (e.g., the 2024 US election), LLM “forecasts” may reflect recall of known outcomes rather than genuine prediction, biasing Brier scores and α downward. The true α on genuinely uncertain events is likely higher than reported here, making our profitability estimates conservative (Appendix G).

Using $\alpha = 0.149$ yields profitability threshold $s > 2(0.15)(0.149) = 4.5\%$ —achievable on thin markets. Inclusive of Polymarket’s 2% taker fees and 50 bps slippage, the practical threshold rises to $\approx 9.5\%$. A paper-trading simulation on the 60 live forecasts produced zero trades on these high-volume markets, correctly validating that the agent abstains from liquid venues. Testing the affirmative claim (profitability on thin markets with $s_m > 15\%$) remains an important direction for future work.

VII. DISCUSSION AND CONCLUSION

Implications for prediction market design.: If the profitability conditions hold under more realistic assumptions (inventory risk, endogenous φ , transaction costs), platforms could integrate forecasting agents as supplementary liquidity providers on thin markets. A layered architecture—LMSR [7] providing baseline liquidity while informed agents tighten spreads—is one candidate design [8].

Scaling dynamics.: As calibration improves, agents could in principle serve markets with narrower spreads. Whether this creates a positive feedback loop (better liquidity $\rightarrow$ more participants $\rightarrow$ better price discovery $\rightarrow$ better training signal) or a negative one (tighter spreads $\rightarrow$ more informed-trader entry $\rightarrow$ higher $\varphi \rightarrow$ losses) depends on the equilibrium, which is not derived here.

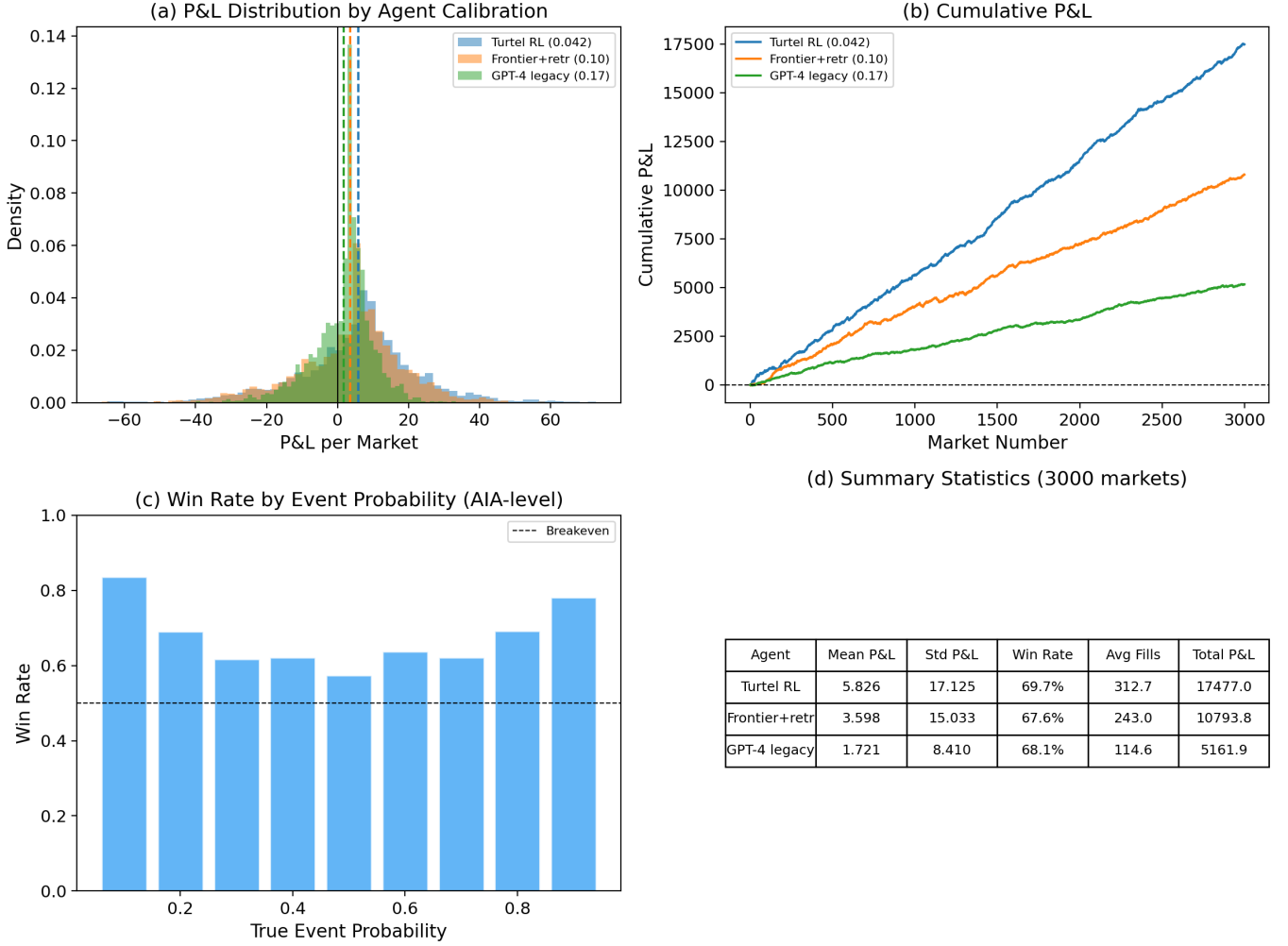


Fig. 2. Monte Carlo Simulation (3,000 Markets per Agent). (a) P&L distributions by calibration level. (b) Cumulative P&L. (c) Win rate by event probability. (d) Summary statistics with 95% CIs.

Transaction costs.: The theoretical framework omits transaction costs. On Polymarket, 2% taker fees and ~ 50 bps slippage raise the profitability condition to $s > 2\varphi\alpha + 2c$ where c is the round-trip cost. With $\varphi = 0.15$, $\alpha = 0.042$, and $c = 2.5\%$, the threshold rises from 1.3% to 6.3%, reducing the improvable market set. All results should be interpreted net of these costs in deployment.

Practical opportunity size.: Fig. 4(c) shows that $\approx 9\%$ of the 98 sampled Polymarket markets have spreads wide enough for agent improvement at current calibration levels. While this is a minority, the absolute count—given thousands of Polymarket markets—represents a meaningful opportunity. Importantly, 63% of active short-term markets have zero 24-hour volume [3]; these ultra-thin markets are not captured in our 98-market sample and likely have wider spreads, expanding the addressable set.

Inventory risk.: The static per-fill model captures expected profit but not tail risk. A market maker holding shares that resolve to 0 or 1 faces catastrophic loss on individual po-

sitions. This is the most important omission in the model. The Monte Carlo P&L standard deviations ($\$8$ – $\$17$ per market) imply per-market Sharpe ratios of ≈ 0.3 – 0.5 . Under a simplified Kelly framework, the probability of drawdown exceeding fraction d of bankroll satisfies $\Pr(\text{drawdown} > d) \leq e^{-2d^2n \cdot \text{SR}^2}$ for n independent markets. At $\text{SR} = 0.35$ and $n = 100$ markets, a 50% drawdown has probability < 0.01 —but this assumes independence, which correlated model errors violate (Appendix G). Fractional Kelly sizing ($f \leq 0.25$; Appendix A) provides additional protection. Integrating inventory dynamics with the spread optimization is the most important direction for future work.

Limitations and open problems.: This paper derives single-agent foundations under simplifying assumptions; several define a natural research frontier:

- 1) **Static φ .** Proposition IV.7 bounds degradation under exogenous φ increase, but in practice narrow spreads attract informed traders, making φ endogenous to s . The equilibrium $\varphi^*(s^*)$ is not derived here; the profitability condition is

therefore necessary but may not be sufficient in a dynamic setting.

- 2) **Single-agent model.** Competition between calibrated agents compresses spreads; Nash equilibrium analysis under n symmetric agents remains open.
- 3) **Correlated errors.** The spread decomposition holds per agent, but aggregate market quality depends on the error covariance structure.
- 4) **Binary outcomes.** Extension to multi-outcome markets requires integration with cost-function market makers [9].
- 5) **Empirical scope.** Longitudinal deployment on thin markets ($s_m > 15\%$) is needed to validate the long-tail thesis; the paper-trading exercise on 60 live forecasts across high-volume markets correctly produced zero fills, confirming that the agent appropriately abstains from liquid venues, but the affirmative claim (profitability on thin markets) remains to be tested in deployment.

Conclusion.: This paper derives a spread decomposition $s^* = 1/\kappa + 2\varphi\alpha$ linking forecast calibration error to market-making profitability on prediction markets. The profitability condition $s > 2\varphi\alpha$ holds for any positive fill-rate function; the optimal spread depends on the specific functional form but preserves the additive microstructure + adverse-selection structure. A comparative static shows that wider markets tolerate higher α , suggesting a role for agents on illiquid long-tail venues—though this result rests on the absence of inventory risk and competition in the model.

The most practically important finding is empirical: market-alignment error (MAD) underestimates true α by an order of magnitude on resolved outcomes. This calibration gap has implications beyond market making, for any system that treats agreement with market prices as evidence of forecast quality.

Ethics statement.: This work uses publicly available market data. No private data, human subjects, or real-money trades are involved. Three failure modes (correlated errors, retrieval poisoning, reflexivity) are analyzed in Appendix G.

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APPENDIX

Proof of Lemma III.4. Jensen’s inequality applied to $|\cdot|$ within each bin b : $|\bar{\varepsilon}_b| \leq |\varepsilon|_b$. Weighting by bin fractions n_b/N :

$$\text{ECE}_B = \sum_b \frac{n_b}{N} |\bar{\varepsilon}_b| \leq \sum_b \frac{n_b}{N} |\varepsilon|_b = \alpha.$$

As $B \rightarrow \infty$, each bin contains at most one sample, collapsing the inequality. Published ECE values (typically 10-bin) thus provide an optimistic lower bound on α . $\square$

Proof of Proposition IV.1. The agent buys a “Yes” share at price b . The share pays 1 if $o = 1$ and 0 if $o = 0$. Expected payoff is $\mathbb{E}[o] = p^*$, so expected profit is $p^* - b = p^* - \hat{p} + s/2$. The ask case is symmetric. $\square$

Proof of Theorem IV.2. With probability φ , the counterparty is informed and selects the unfavorable side: profit is $s/2 - |\hat{p} - p^*|$. With probability $1 - \varphi$, the counterparty is uninformed: profit is $s/2$. The expected profit per fill is $s/2 - \varphi |\hat{p} - p^*|$. Multiplying by $F(s)$ gives the profit rate. Taking expectations yields (3). $\square$

Proof of Theorem IV.4. $\mathbb{E}[\Pi(s)] = \lambda e^{-\kappa s} (s/2 - \varphi\alpha)$. Differentiating:

$$\frac{d\mathbb{E}[\Pi]}{ds} = \lambda e^{-\kappa s} \left[\frac{1}{2} - \kappa \left(\frac{s}{2} - \varphi\alpha \right) \right] = 0.$$

Solving: $s^* = 1/\kappa + 2\varphi\alpha$. The second-order condition $d^2\Pi/ds^2 < 0$ is verified by inspection. $\square$

Conservative adverse selection bound.: In a refined model where informed traders only trade when $|\hat{p} - p^*| > s/2$, the adverse selection loss becomes $L(s) = \varphi \cdot \mathbb{E}[\max(0, |\hat{p} - p^*| - s/2)] < \varphi \alpha$. The spread screens out marginal informed traders, making $s > 2\varphi\alpha$ sufficient but not necessary.

The agent’s forecasting pipeline follows [4], [5]: (1) query decomposition; (2) retrieval and filtering [20]; (3) LLM reasoning with RL-tuned models [6]; (4) ensemble aggregation via trimmed mean; (5) Platt scaling calibration. Position sizing follows fractional Kelly criterion.

A. Optimal Spread Analysis

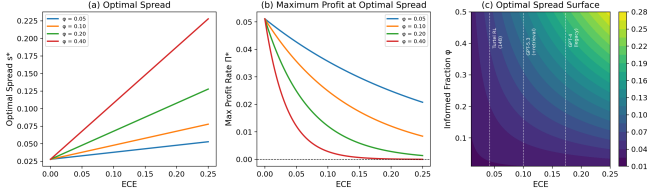


Fig. 3. Optimal Spread Analysis. (a) s^* vs. α . (b) Max profit rate. (c) Spread surface over (α, φ) .

B. Calibration Threshold with Real Data

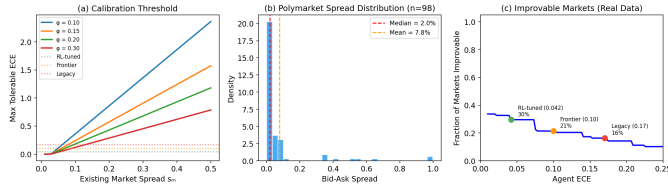


Fig. 4. Calibration Threshold. (a) Max α for improvement. (b) Polymarket spread distribution ($n = 98$). (c) Fraction improvable.

C. Sensitivity Analysis

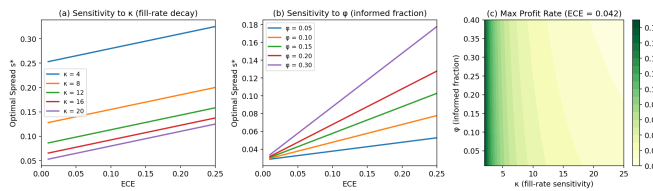


Fig. 5. Sensitivity. (a) s^* vs. α for different κ . (b) For different φ . (c) Profit surface.

D. Empirical κ Calibration

E. Market Quality Impact

F. Calibration–Cost Frontier

G. Backtest Details

The 15 resolved questions span domains (elections, policy, sports, geopolitics) and resolution dates (2024–2026). For well-known events, models likely have outcomes in training data, biasing Brier and ECE downward. The paper-trading

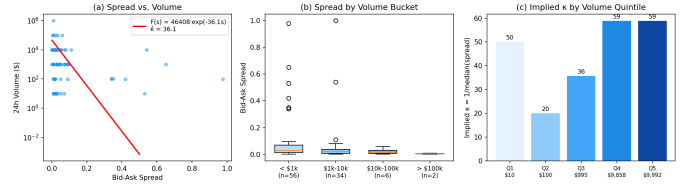


Fig. 6. Empirical κ . (a) Volume vs. spread; fitted $\hat{\kappa} = 36.1$. (b) Spread by volume. (c) Implied κ by quintile.



Fig. 7. Market Quality. (a) Effective spread with/without agent. (b) Price efficiency. (c) Agent P&L vs. thickness.

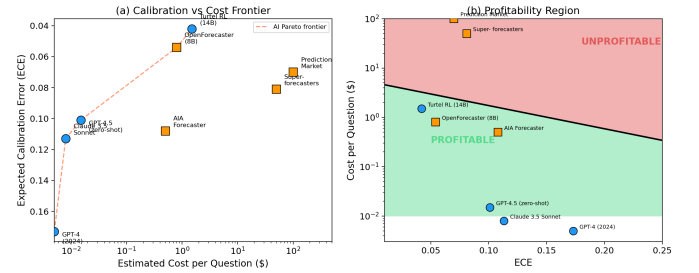


Fig. 8. Calibration–Cost Frontier. (a) ECE vs. cost. (b) Profitability region.

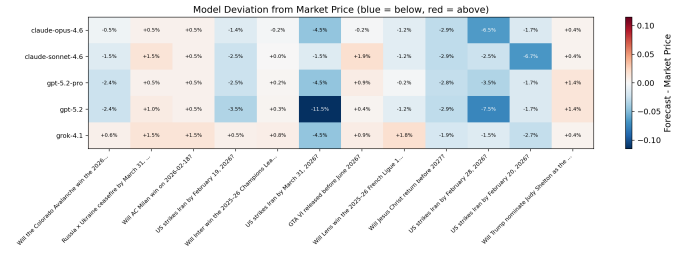


Fig. 9. Model Deviation Heatmap. Geopolitical questions show widest disagreement.

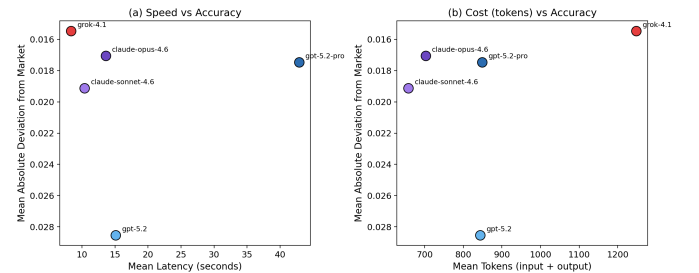


Fig. 10. Speed–Accuracy Frontier. Grok 4.1 is Pareto-optimal (8.3s, MAD 0.015).

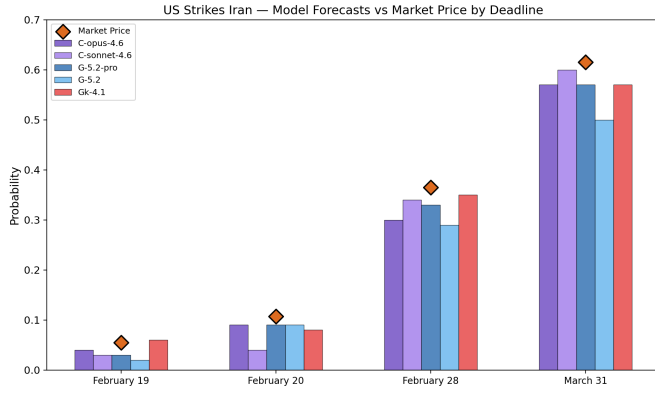


Fig. 11. Iran Strikes Term Structure. Models correctly reproduce monotonic structure but underforecast.

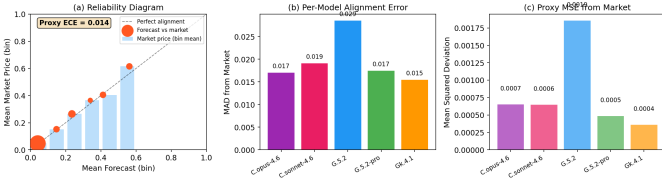


Fig. 12. Proxy Calibration. (a) Reliability diagram. (b) MAD by model. (c) MSD by model.

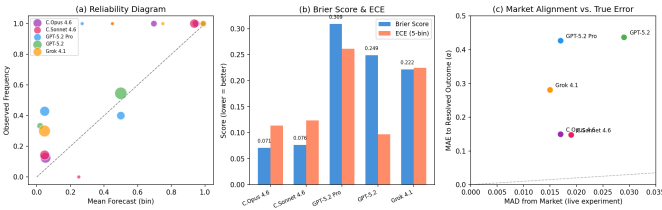


Fig. 13. Resolved-Outcome Calibration. (a) Reliability diagram. (b) Brier/ECE. (c) MAD vs. true α .

simulation used 2% taker fees, $\hat{\kappa} = 36$, 50 bps slippage, and fractional Kelly ($f = 0.25$) with \$10K bankroll, producing zero fills on 60 forecasts across high-volume markets.

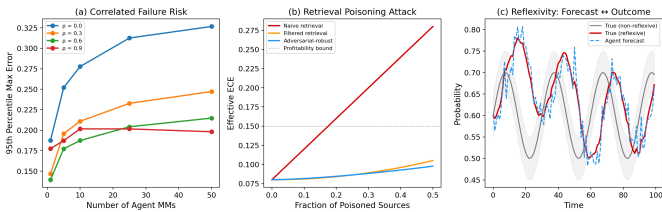


Fig. 14. Failure Modes. (a) Correlated errors. (b) Retrieval poisoning. (c) Reflexivity.

Correlated errors.: Shared training data create correlated errors. At $\rho = 0.9$, diversification is negligible. *Mitigation*: architectural diversity.

Retrieval poisoning.: Naive pipelines suffer catastrophic ECE degradation when 20% of sources are manipulated. *Mitigation*: two-stage filtering [20].

Reflexivity.: Agent forecasts can influence outcomes on endogenous markets. *Mitigation*: restrict to exogenous-outcome markets.

Code and data: anonymized repository (link upon acceptance). Spread data from Polymarket Gamma API (Feb 18, 2026, $n = 98$ markets). Live experiment via OpenRouter (temperature 0.2, max 800 tokens). Resolved-outcome backtest: 15 questions with clear binary resolution.